

GSoC 2026 Project Proposal

Quantum Model Agnostic Meta Learning for Variational Quantum Algorithms for HEP Analysis at the LHC

Organization: ML4SCI

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1 Introduction and Contributor Information

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2 Abstract

The High-Luminosity Large Hadron Collider (HL-LHC) will generate unprecedented volumes of high-dimensional data, requiring scalable and efficient learning paradigms for physics analysis. While classical ML methods have demonstrated success in High Energy Physics (HEP), their training cost and scalability remain limiting factors.

Variational Quantum Algorithms (VQAs) offer a promising alternative through hybrid quantum-classical optimization. However, their trainability is hindered by *barren plateaus* regions in the parameter landscape where the gradient of the cost function vanishes exponentially with system size (McClean et al., 2018):

$$\nabla_{\theta} \mathcal{L}(\theta) \sim \mathcal{O}(2^{-n}),$$

where n is the number of qubits. Poor parameter initialization exacerbates this, leading to slow or non-convergence (McClean et al., 2018; Zhang et al., 2022).

This project proposes the application and extension of **Quantum Model-Agnostic Meta-Learning (Q-MAML)** (Lee et al., 2025) to HEP tasks, building directly on the successful GSoC 2025 implementation. Q-MAML adapts the classical MAML framework (Finn et al., 2017) to quantum settings: a classical neural network (“Learner” h_W) predicts effective initialization parameters θ for parameterized quantum circuits (PQCs) by solving the meta-objective over a distribution of tasks $p(\mathcal{T})$:

$$\arg \min_W \sum_{T_i \sim p(\mathcal{T})} l_{T_i}(g(h_W(\phi_i))),$$

where $g(\theta)$ is the PQC and ϕ_i is the task embedding. Unlike standard MAML, which requires a second-order gradient through the inner loop, Q-MAML uses a first-order approximation via the differentiable chain rule through the quantum expectation value:

$$\frac{dl_T}{dW} = \frac{dl_T}{d\langle O \rangle} \cdot \frac{d\langle O \rangle}{d\theta} \cdot \frac{d\theta}{dW}.$$

This yields moderate gradient norms (Lee et al., Figs. 6–8) and 2–3 $\times$ faster convergence while avoiding barren plateaus (cf. Gaussian initializations of Zhang et al., 2022).

The 2025 GSoC project already demonstrated Q-MAML on quark–gluon jet tagging and Higgs classification using ResNet-18 embeddings as task vectors, achieving up to 78% accuracy with significantly faster convergence than random or Gaussian initializations. This proposal extends it to additional LHC tasks, multi-task scaling, hardware execution on IBM Quantum devices, and rigorous benchmarking against classical ML baselines and standard VQAs, directly addressing the project goal of Q-MAML for HEP tasks.

3 Technical Details

3.1 Q-MAML Framework for Variational Quantum Algorithms

Variational Quantum Algorithms minimize a cost function of the form

$$E(\theta) = \langle \psi(\theta) | H | \psi(\theta) \rangle \quad \text{or} \quad \mathcal{L}_{\text{class}}(\theta),$$

where $\psi(\theta) = g(\theta)|0\rangle$ is the variational state prepared by the PQC $g(\theta)$.

Q-MAML (Lee et al., 2025) builds on the model-agnostic meta-learning paradigm of Finn et al. (2017), which trains initial parameters θ such that a small number of gradient steps on a new task T_i yields rapid adaptation. For quantum circuits, Lee et al. replace the inner loop Hessian with a first-order chain rule differentiation through the quantum observable, enabling scalable training of a classical *Learner* h_W that maps task embeddings ϕ (Hamiltonian vectors or ResNet-18 features in the HEP extension) to PQC parameters:

$$\theta = h_W(\phi).$$

The meta-objective becomes

$$\arg \min_W \sum_{T_i \sim p(T)} l_{T_i}(g(h_W(\phi_i))).$$

Pre-training Phase: The Learner is trained across a distribution of tasks (Hamiltonians or HEP datasets), producing initializations that generalize across the combined task landscape.

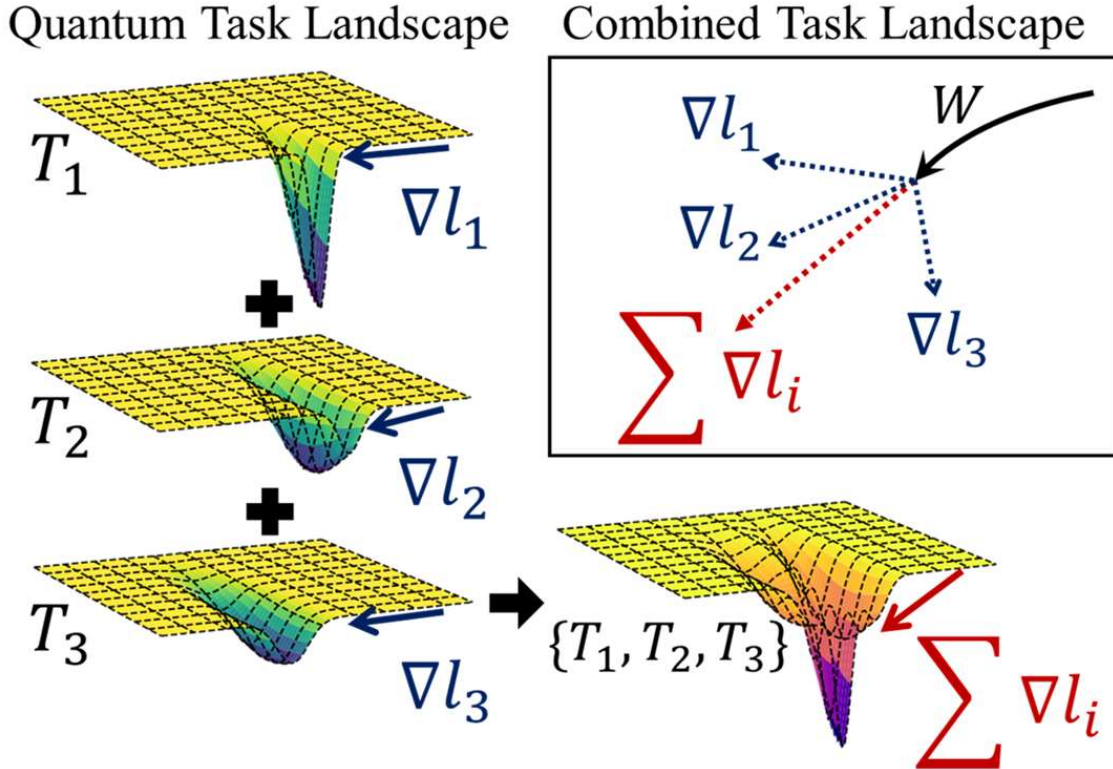


Figure 1: Illustration of the multiple individual optimization landscapes ($\nabla l_1, \nabla l_2, \nabla l_3$) combine to form a single landscape ($\sum \nabla l_i$). The model trained using the combined landscape can capture the general properties of various tasks which results in the improvement of convergence in the optimization as in the MAML.

Adaptation Phase: For a target task T^* , initialize $\theta = h_{W^*}(\phi)$ and refine with a classical optimizer while freezing the Learner.

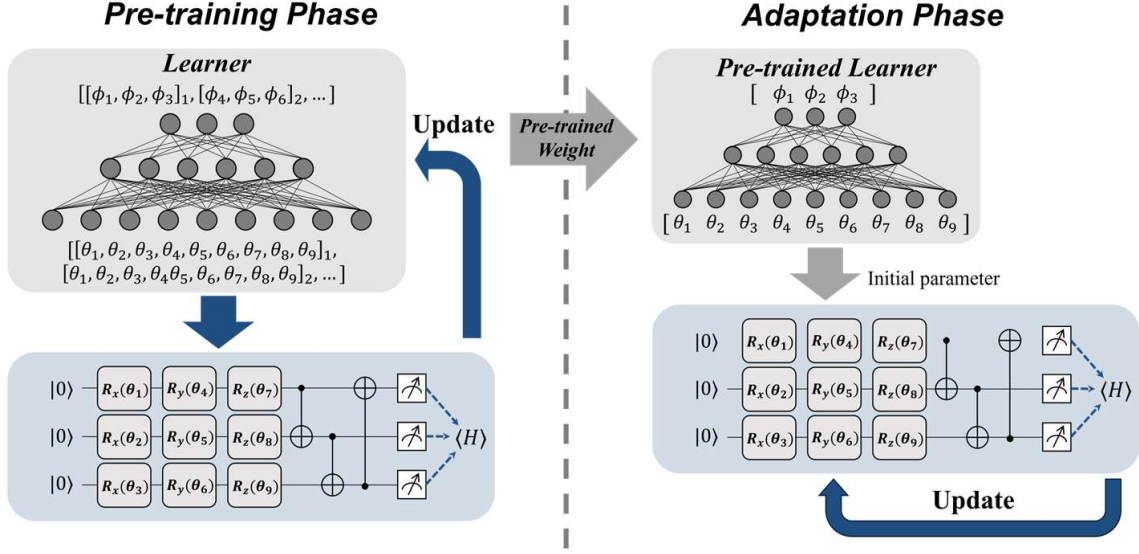


Figure 2: The pre-training phase includes an update of the *Learner* while the adaptation phase trains PQC on specific tasks with parameter initialized using pre-trained *Learner*.

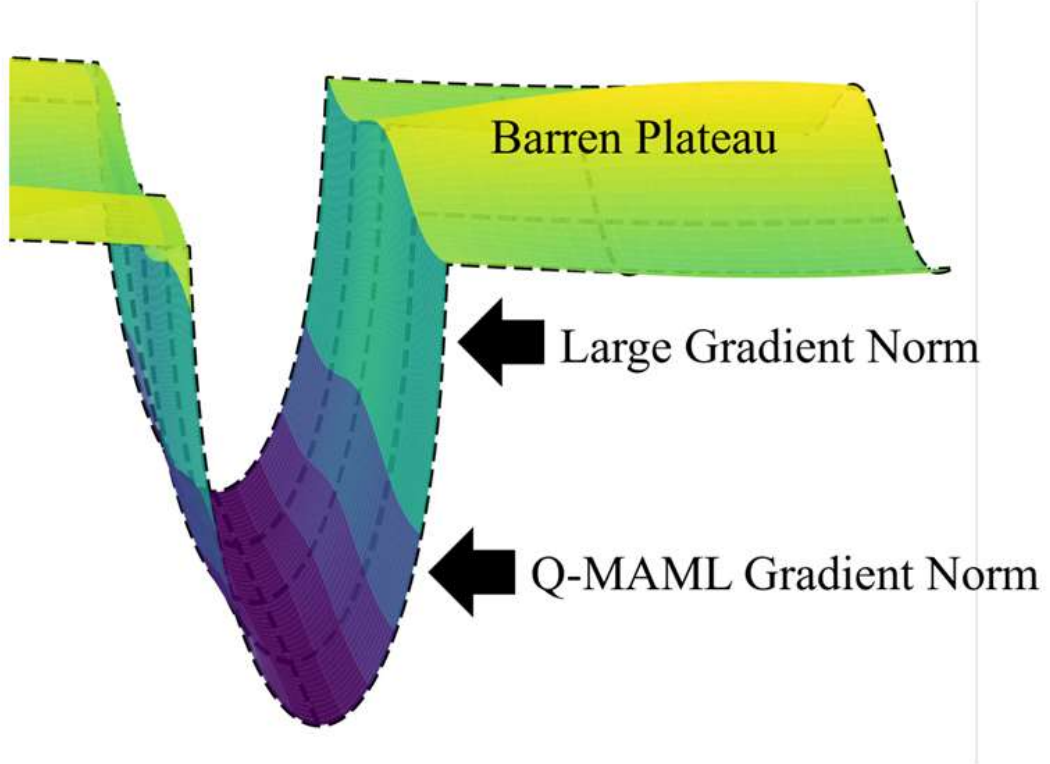


Figure 3: The “Barren Plateau” region is depicted in yellow. The green and blue regions indicate areas with “Large Gradient Norm” and “Q-MAML Gradient Norm”.

Empirical results on Heisenberg XYZ ($\hat{H}_{XYZ} = -\sum(J_x \sigma_n^x \sigma_{n+1}^x + \dots)$) and molecular Hamiltonians, as well as the 2025 HEP extension, show:

- Task-specific patterns in initializations with broader variance/mean than uniform/Gaussian methods (Lee et al., Fig. 5).
- Gradient norms converge to moderate values, yielding $2\text{--}3\times$ faster convergence (Lee et al., Figs. 4, 6–7).
- Initialization lies closer to the global minimum (Lee et al., Fig. 8).

3.2 Related Meta-Learning for Quantum Optimization

Complementary to Q-MAML, Verdon et al. (2019) demonstrated classical recurrent neural networks (RNNs) as meta-optimizers for quantum neural networks.

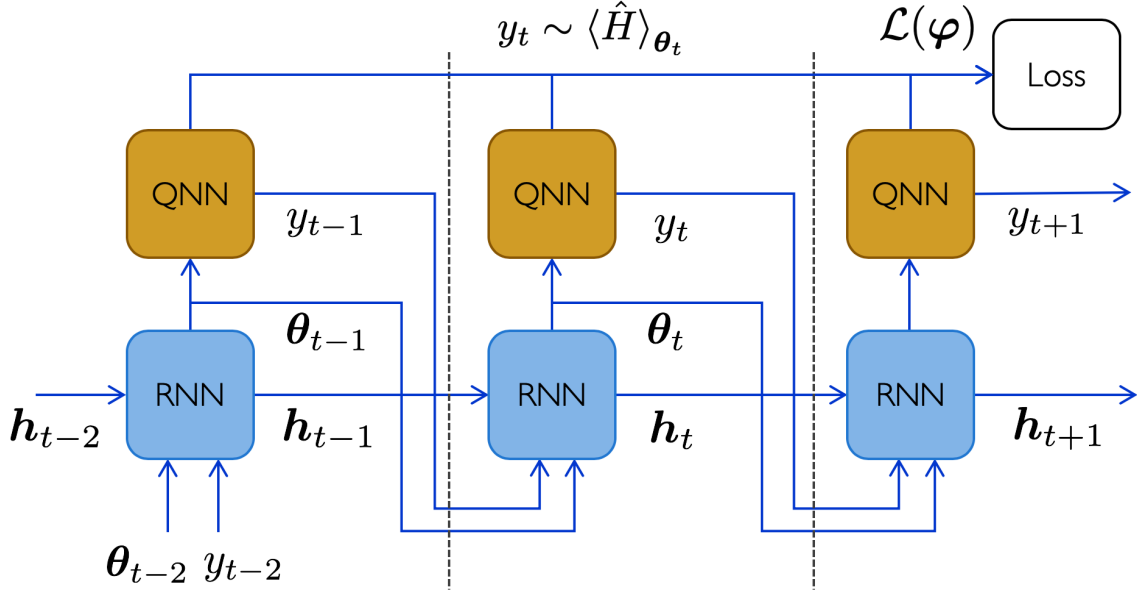


Figure 4: Unrolled temporal quantum-classical computational graph for the meta-learning optimization of the recurrent neural network (RNN) optimizer and a quantum neural network (QNN). The memory of the optimizer is encoded in the hidden state of the RNN h_t . Represent the flow of data used to evaluate the meta-learning loss function. This meta loss function L is a functional of the history of expectation value estimate samples $y = \{y_t\}_{t=1}^T$, and is thus indirectly dependent on the RNN parameters ϕ . Backpropagating from the meta-loss node to the RNN’s necessitates gradients to pass through the QNN.

3.3 Q-MAML for HEP Task Distributions

We generalize Q-MAML to data-driven HEP tasks (quark/gluon jets, Higgs vs. background), where each task T_i is a dataset and ϕ_i is a task embedding (physics features or CNN/ResNet embeddings). The meta-objective is

$$\arg \min_W \sum_{T_i \sim p(\mathcal{T}_{\text{HEP}})} \mathcal{L}_{\text{class}}(g(h_W(\phi_i))).$$

This enables rapid few shot adaptation to new LHC processes, critical for rare signal searches in the HL-LHC era and complements related meta-learning efforts for quantum optimization (Verdon et al., 2019).

3.4 Data and Model Architecture

- **Datasets:** Synthetic HEP data, CMS Open Data, simulated LHC events (Pythia + Delphes); extending the 2025 quark–gluon ($125 \times 125 \times 3$ images) and Higgs datasets.
- **Feature Extractor:** CNN/ResNet embeddings $\rightarrow$ task vector ϕ .
- **Meta Learner:** MLP (input: task embedding; output: PQC parameters).
- **Quantum Model:** Variational Quantum Classifier (6–12 qubits, StronglyEntanglingLayers or hardware-efficient ansätze).
- **Cost Function:** Binary/cross-entropy loss on classification.

3.5 Implementation Requirements

- PennyLane (quantum simulation & differentiable circuits).
- PyTorch (meta-learning & Learner training).
- IBM Qiskit Runtime (hardware execution on real quantum devices).

3.6 Past Results

Initialization	Accuracy (%)	Convergence (epochs)	Variance
Q-MAML (Learner)	78.4	Fast (~ 600)	Low
Gaussian	68.2	Medium (~ 1400)	High
Uniform	65.9	Slow (~ 1800)	High
Zero	64.1	Slow (~ 2000)	High

4 Project Deliverables

I will start with a thorough literature review to understand current quantum meta-learning techniques and identify potential improvements for HEP tasks. After preprocessing relevant LHC datasets for training, I will try different Q-MAML configurations, evaluated using metrics such as convergence speed, accuracy, and gradient norms. Then compare to standard initialization methods and classical ML baselines. Every step will be heavily documented for reproducibility and guiding future work.

- **Literature Review:**

- A review of Q-MAML (Lee et al., 2025), classical MAML (Finn et al., 2017), and barren plateau mitigation techniques (McClellan et al., 2018; Zhang et al., 2022).
- Analyzing quantum meta-learning approaches and related works on variational quantum algorithms for HEP.
- Identifying gaps in current HEP quantum ML methods and understanding how meta-learned initialization can address these challenges.

- **Dataset Preparation:**

- Preprocessing relevant gravitational lensing datasets (quark/gluon jet tagging, Higgs vs. background classification) from CMS Open Data and simulated LHC events.
- Applying standard data processing techniques such as feature extraction using CNN/ResNet embeddings, normalization, and augmentation.

- **Experimenting with Different Q-MAML Architectures:**

- Exploring various model architectures including Variational Quantum Classifiers with 6–12 qubits.
- Testing StronglyEntanglingLayers and hardware-efficient ansätze.
- Evaluating both unconditional and task conditioned setups to capture complex HEP phenomena.

- **Evaluation Metrics:**

- Accuracy, convergence speed (number of epochs/iterations), and gradient norm analysis.
- Comparison with standard initializations (Zero, Uniform, Gaussian, π).
- Benchmarking against classical ML baselines (XGBoost, ResNet) and standard VQAs.
- Quantum resource usage (circuit evaluations and hardware runtime).

- **Documentation:**

- Compiling the final results into a research report (or paper draft) detailing methodologies, evaluation, and conclusions.

4.1 4.2 Timeline for Deliverables

- **Community Bonding Period:** During this period, I will engage with my mentors and the ML4SCI community. Review past contributors' work (especially the 2025 GSoC Q-MAML project) and deepen my understanding of Q-MAML, barren plateaus, and HEP specific quantum challenges. I will also set up the development environment and reproduce key results from Lee et al. (2025) and the 2025 GSoC codebase.
- **Week 1:** Conduct a literature review on Q-MAML, MAML, and barren plateau mitigation techniques. Study existing HEP quantum ML pipelines and identify integration points for meta-learning.
- **Week 2:** Preprocess HEP datasets (quark/gluon jets and Higgs classification). Implement feature extraction using CNN/ResNet embeddings and setup the data pipeline for task embedding generation.
- **Week 3:** Implement baseline VQAs (6–12 qubit classifiers) with standard initializations. Establish evaluation framework (accuracy, convergence speed, gradient norms) and run initial benchmarks.
- **Week 4:** Fully implement the Q-MAML framework — including the classical Learner network, pre-training phase, and adaptation phase. Begin training the meta-learner on the task distribution.
- **Week 5:** Complete multi-task pre-training of the Learner. Perform adaptation experiments on individual HEP tasks and compare convergence with standard initialization methods.
- **Week 6:** Extend to multi-task scaling and improved task embeddings. Conduct ablation studies on different embedding strategies. Prepare mid-term report and first blog post.
- **Week 7:** Integrate hardware execution using IBM Qiskit Runtime. Run Q-MAML on real quantum devices and compare simulation vs. hardware performance.

- **Week 8:** Perform comprehensive benchmarking against classical ML models (XGBoost, ResNet) and standard VQAs. Analyze gradient norms and convergence improvements.
- **Week 9:** Optimize the framework for efficiency (fewer quantum circuit evaluations). Explore conditional setups and refine the meta-learner architecture based on results.
- **Week 10:** Conduct final large scale experiments and ablation studies. Prepare the second blog post detailing hardware results and optimizations.
- **Week 11:** Complete the research report (or paper draft), clean up the codebase and compile all results.
- **Week 12:** Finalize all deliverables and prepare for the final evaluation. Discuss potential post-GSoC collaborations with mentors.

5 Potential Impact

This project could directly address the computational challenges of HL-LHC data analysis by providing faster converging, resource efficient quantum models via meta-learned initializations. It has the potential to demonstrate the practical value of classical meta-learning (Finn et al., 2017; Verdon et al., 2019) for quantum optimization in HEP and contribute reusable tools to the ML4SCI community.

6 Why ML4SCI

My research areas are quantum optimization, variational algorithms, mathematical computing and machine learning. ML4SCI combines these in one, making it a great fit for me. I want to work on problems where machine learning helps quantum models and ML4SCI (specifically QML-HEP) focuses on that.

Since it is part of CERN, it offers access to real world scientific challenges and datasets. I have worked on deep learning models, including diffusion models and transformers, and explored ways to optimize inference using tools like TensorRT and CUDA. I am particularly interested in extending these ideas to hybrid quantum-classical settings.

My goal is to do a PhD, and working in ML4SCI would help me build the right experience. By contributing to ML4SCI, I can work on open research and collaborate with others who share my interest in quantum computing and machine learning.

7 Past Experience

- **Quantum Inspired Active Learning for Materials Discovery** (NQComp 2026)
Wrote a paper on a quantum inspired active learning framework using amplitude encoding in Hilbert space and multi-observable uncertainty aggregation with non-commuting Hermitian operators for material science. Derived covariance aware acquisition functions modeling cross-property structure–property correlations. Across five independent trials and nine baselines (QBC, EI, BADGE, CoreSet, GP based methods), achieved 2.6–10.4% improvements in test R^2 and 25–35% reduction in sample complexity ($p < 0.01$). Complexity analysis shows scalable $O(NKd)$ performance under structured operators.
- **Lattice QCD and Critical Slowing Down**
Supervisor: Pablo Morande (University of Cambridge)
Implemented Hybrid Monte Carlo (HMC) and Metropolis–Hastings algorithms for lattice field simulations to analyze autocorrelation scaling and critical slowing through discretization of the harmonic oscillator.

- **Research Intern - JuliaSmoothOptimizers(JSO)**
Supervisor: Dominique Orban (Polytechnique Montreal)
Developing a structured pipeline to translate classical optimization problems into ADNLP-Models and JuMP formulations for integration into OptimizationProblems.jl.
- **Bayesian Modeling – sktime**
Supervisor: Dr. Franz Kiraly (GCOS)
Working on implementing Bayesian probabilistic estimators within the skpro ecosystem. Developing posterior predictive models, integrating uncertainty quantification and proper scoring rules (e.g. log-score, CRPS).
- **LFX Mentee – Linux Kernel**
Contributed patches across filesystems(ext4, ntfs3, hpfs), memory management and networking subsystems. Fixed use-after-free vulnerabilities and improved kernel build and error handling mechanisms.

8 Commitment

I am completely free in the Summer and am prepared to give my full attention to the project.

In case of unavoidable circumstances leading to a temporary reduction in working hours, I will ensure to compensate for the time in the following weeks. I will communicate any such situations with my mentors well in advance to ensure smooth project progress.

For regular updates and discussions, I will be available via Teams, Zoom, and Google Meet in the Indian Standard Time Zone (GMT +5:30)

References

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- GSoC 2025 Q-MAML Repository: <https://github.com/arnavsinghal09/GSoC-ML4Sci-QMAML>
- HL-LHC: <https://home.cern/science/accelerators/high-luminosity-lhc>
- LHC: <https://home.cern/science/accelerators/large-hadron-collider>
- PennyLane: <https://pennylane.ai/>